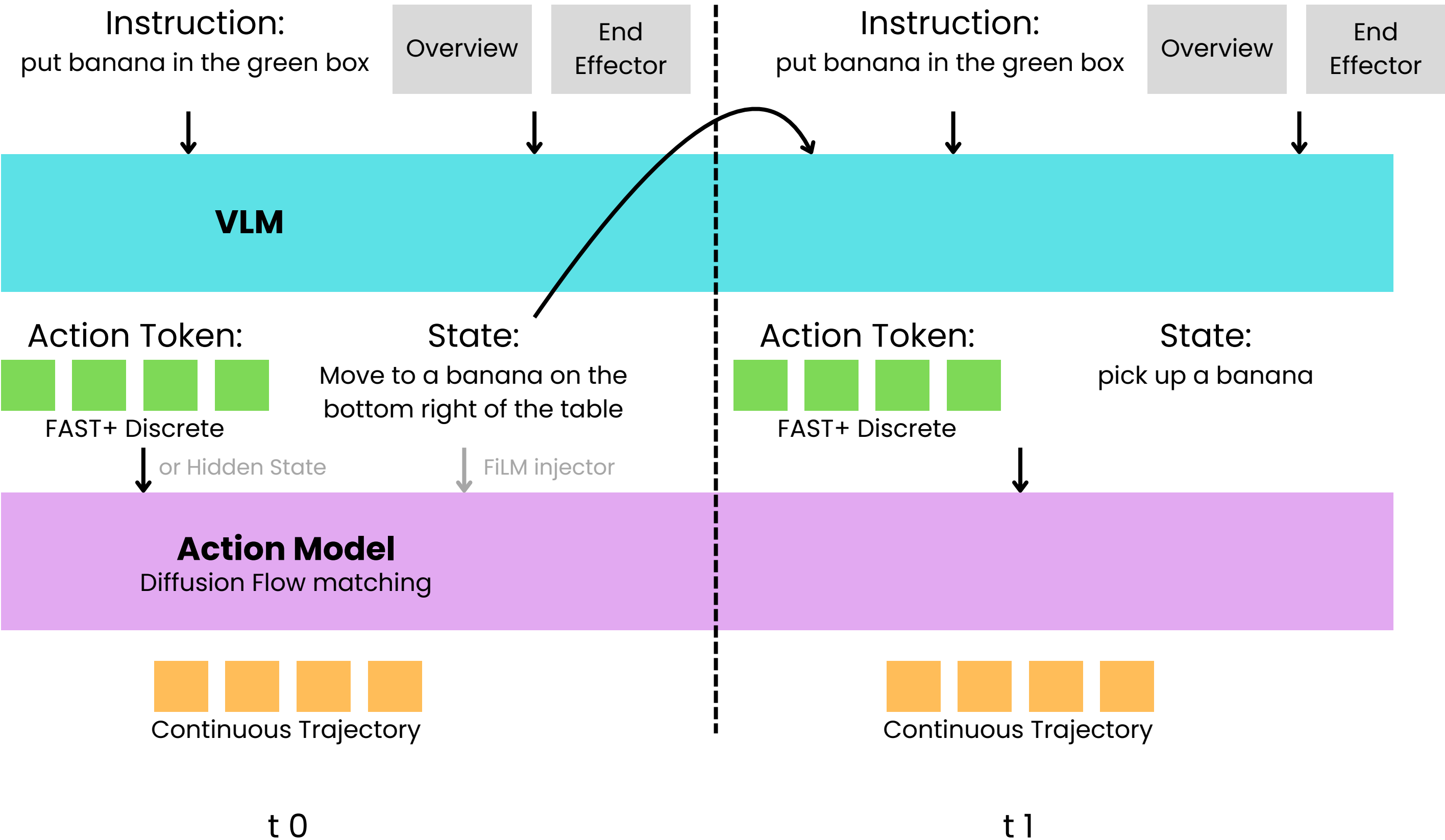


Normal VLA architecture



There are 3 Type of States:

- Text State**
- Special Token
- Continuous State*

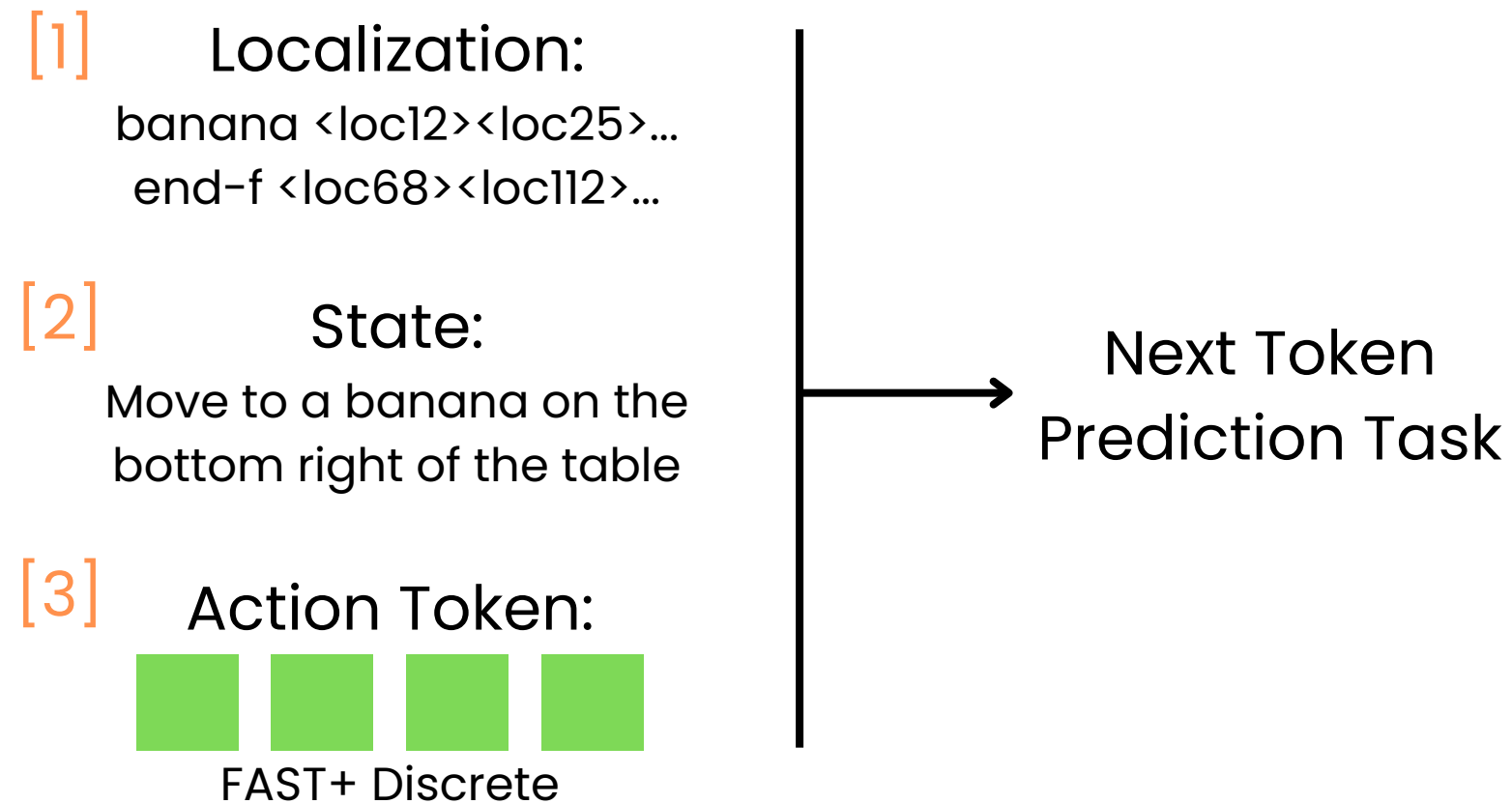
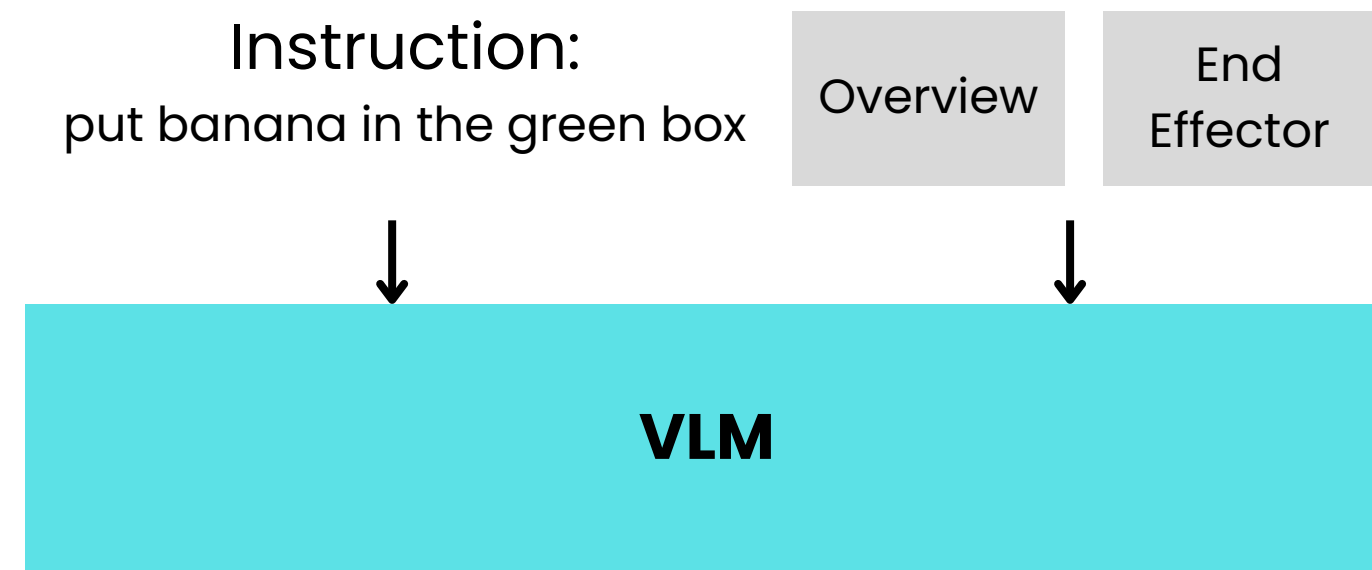
Action Tokenizer

- FAST+ (SOTA)**
- Customized*

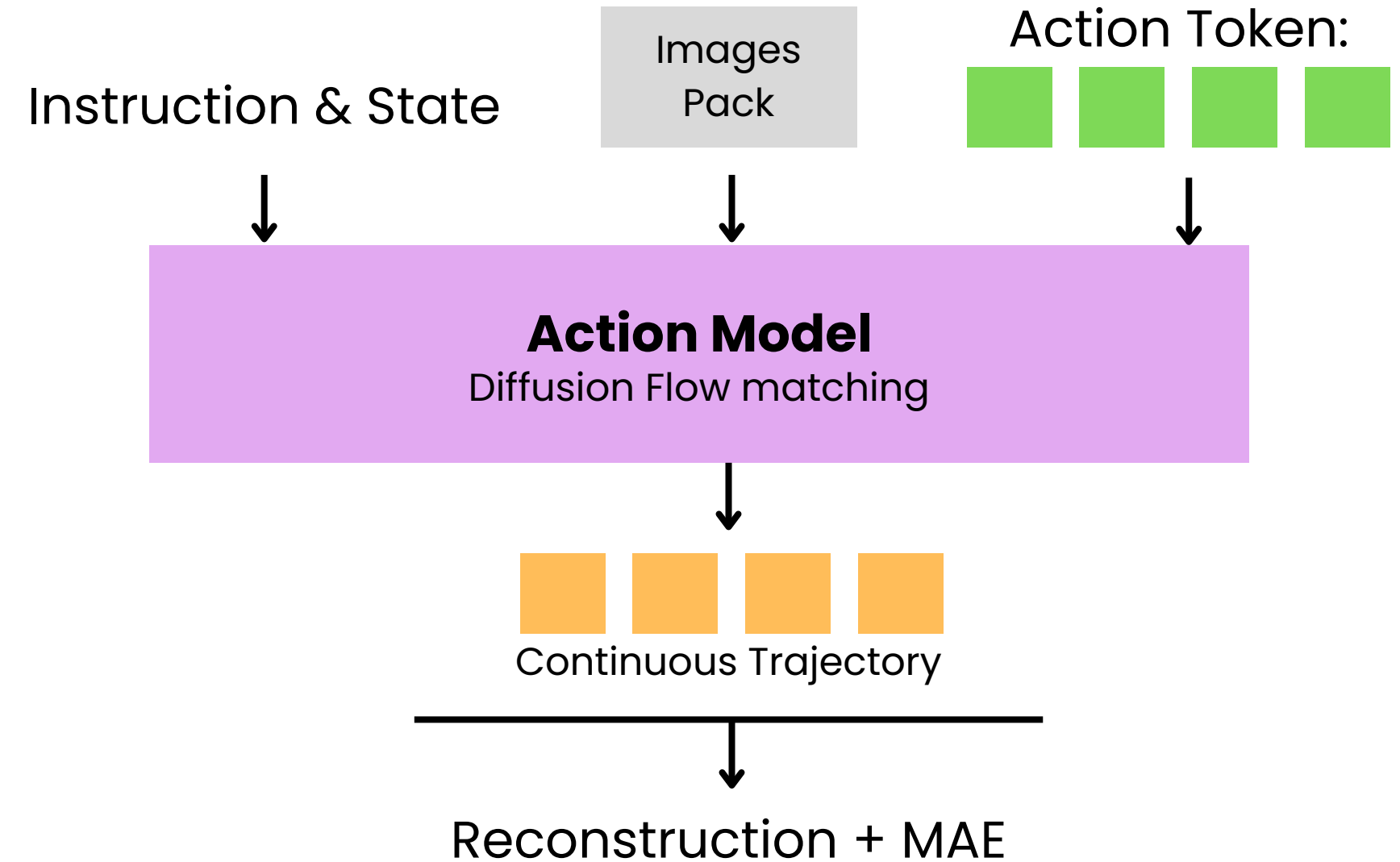
Note:
NVIDIA use = *
Best Practice = **

Normal Training Process (Pre Training Stage)

The Foundation Model



Action Model



References (Overall)

Development Timeline and Best Current Best Practice

[Vision-Language-Action Models: Concepts, Progress, Applications and Challenges](#)

FAST universal action Tokenizer

[FAST: Efficient Action Tokenization for Vision-Language-Action Models](#)

Pre-Training Best Practice

[Knowledge Insulating Vision-Language-Action Models: Train Fast, Run Fast, Generalize Better](#)

Common Backbone Architecture (multi encoder SigLIP + DiNO)

[OpenVLA: An Open-Source Vision-Language-Action Model](#)

Object Localization in LLM

[Learning to Localize Objects Improves Spatial Reasoning in Visual-LLMs](#)

Diffusion base Action Expert

[Diffusion-VLA: Generalizable and Interpretable Robot Foundation Model via Self-Generated Reasoning](#)

NVIDIA practice

[GR00T N1: An Open Foundation Model for Generalist Humanoid Robots](#)

Our Architecture

LLMs that learn to control robot directly

